Southern Nuclear

Regulatory Affairs

3535 Colonnade Parkway
Birmingham, AL 35243

Tel. 205 992 5000

May 23, 2025

Docket No.: 50-366 NL-25-0200

U. S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, D. C. 20555-0001

Edwin I. Hatch Nuclear Plant
Licensee Event Report 2025-002-00
2A Emergency Diesel Generator Inoperable Due to Inadvertent Overspeed Trip

Ladies and Gentlemen:

In accordance with the requirements of 10 CFR 50.73(a)(2)(i)(B) and 10 CFR 50.73(a)(2)(v)(D), Southern Nuclear Operating Company hereby submits the enclosed Licensee Event Report.

This letter contains no NRC commitments. If you have any questions, please contact the Hatch Licensing Manager, Jimmy Collins, at 912.453.2342.

Respectfully submitted,

Jamie Coleman

Jamie M. Coleman
Regulatory Affairs Director
Southern Nuclear Operating Co., Inc.

JMC/JMH

Enclosure: LER 2025-002-00

Cc: Regional Administrator, Region II
NRR Project Manager – Hatch
Senior Resident Inspector – Hatch
RTYPE: CHA02.004

Edwin I. Hatch Nuclear Plant
Licensee Event Report 2025-002-00

2A Emergency Diesel Generator Inoperable Due to Inadvertent Overspeed Trip

Enclosure

LER 2025-002-00

NRC FORM 366 (04-02-2024) U.S. NUCLEAR REGULATORY COMMISSION

APPROVED BY OMB: NO. 3150-0104 EXPIRES: 04/30/2027

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name		2. Docket Number	3. Page
Edwin I. Hatch Nuclear Plant Unit 2	■ 050 / ☐ 052	00366	1 OF 3

4. Title

2A Emergency Diesel Generator Inoperable Due to Inadvertent Overspeed Trip

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	Docket Number
03	26	2025	2025	- 002 -	00	05	23	2025		☐ 050
										☐ 052

9. Operating Mode	10. Power Level
1	022

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20		10 CFR Part 50			
	☐ 20.2203(a)(2)(vi)		☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	**10 CFR Part 21**	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	**10 CFR Part 73**	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☑ 50.73(a)(2)(i)(B)	☑ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Carl James Collins - Licensing Manager	912-453-2342

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	EK	12	H260	Y					

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☑ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

The 2A Emergency Diesel Generator (EDG) unexpectedly tripped during a test on 2/13/2025, due to its emergency overspeed switch (EOS) falsely signaling an overspeed condition. An initial investigation and operability review found no firm evidence suggesting that the 2A EDG was unable to perform its safety function prior to the trip. Subsequently, the EOS was replaced, the 2A EDG successfully completed its operability surveillance on 2/15/2025, and the original EOS was sent off for failure analysis.

On 3/26/2025, at 14:53 EDT, Unit 2 was operating at approximately 22 percent power in MODE 1 when results from the failure analysis of the EOS for the 2A EDG were received. Contrary to the findings of the initial investigation and operability review, the analysis revealed that the condition experienced with the EOS during the 2/13/2025 test most likely caused the 2A EDG to be inoperable longer than allowed by Technical Specifications (TS). Additionally, during the time period the 2A EDG was inoperable, the 1B EDG was tagged out in support of planned maintenance for the standby plant service water pump resulting in two EDGs being inoperable for Unit 2 longer than allowed by TS. Having two EDGs inoperable for the same unit resulted in an event or condition that could have prevented the fulfillment of a safety function needed to mitigate the consequences of an accident.

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Edwin I. Hatch Nuclear Plant Unit 2	■ 050 / ☐ 052	00366	2025	002	00

NARRATIVE

EVENT DESCRIPTION

The 2A Emergency Diesel Generator (EDG) (EIIS: DG) unexpectedly tripped during a test on 2/13/2025, due to its emergency overspeed switch (EOS) falsely signaling an overspeed condition. An initial investigation and operability review found no firm evidence suggesting that the 2A EDG was unable to perform its safety function prior to the trip. Subsequently, the EOS was replaced, the 2A EDG successfully completed its operability surveillance on 2/15/2025, and the original EOS was sent off for failure analysis.

On 3/26/2025, at 14:53 EDT, Unit 2 was operating at approximately 22 percent power in MODE 1 when results from the failure analysis of the EOS for the 2A EDG were received. Contrary to the findings of the initial investigation and operability review, the analysis revealed that the condition experienced with the EOS during the 2/13/2025 test most likely caused the 2A EDG to be inoperable from 6/6/2024, the date the EOS was installed, to when the 2A EDG successfully completed its operability surveillance on 2/15/2025, which is longer than allowed by Technical Specifications (TS) 3.8.1.B.

Additionally, while reviewing the 2A EDG EOS trip event, it was discovered that the 1B EDG (EIIS: DG) was tagged out for eighteen hours and forty-nine minutes in support of planned maintenance for the standby plant service water pump on 6/21/2024, during the same time the 2A EDG was inoperable. The overlapping time of the inoperabilities of the 2A EDG and the 1B EDG resulted in two EDGs being inoperable for Unit 2 longer than allowed by TS 3.8.1.F. Having two EDGs inoperable for the same unit resulted in an event or condition that could have prevented the fulfillment of a safety function needed to mitigate the consequences of an accident.

FAILED COMPONENTS

Name: Emergency Overspeed Switch (EOS)
Master Parts List Number: N/A; the EOS is a subcomponent of the 2A EDG (2R43S001A)
Manufacturer: Honeywell
Model: BZE6-RQ
Type: Limit Switch

EVENT CAUSE ANALYSIS

Analysis showed that a loose retaining nut on the EOS impacted switch operation and caused the EOS to falsely signal an overspeed condition during the 2/13/2025 test. The EOS installed near the actuation point also contributed to the inadvertent trip of the 2A EDG.

SAFETY ASSESSMENT

There were no actual safety consequences as a result of these events. However, while the 2A EDG was inoperable, both the 2C EDG and the 1B swing EDG remained operable for Unit 2, except when TS surveillances were performed and when the 1B EDG was required to be tagged out for eighteen hours and forty-nine minutes on 6/21/24 in support of planned maintenance for the standby plant service water pump. Due to the 2A EDG being inoperable longer than allowed by TS and the overlapping time of the inoperabilities of the 2A EDG and the 1B EDG, resulting in two EDGs being inoperable for Unit 2 longer than allowed by TS, the completion times of TS 3.8.1.B and TS 3.8.1.F were not met. Additionally, the completion time associated with TS 3.8.1.H to be in MODE 3 in 12 hours was also not met. Therefore, these events are reportable as conditions prohibited by TS per 10 CFR 50.73(a)(2)(i)(B). Additionally, having the 2A EDG and the 1B EDG simultaneously inoperable on the same unit is a condition that could have prevented the fulfillment of a safety function needed to mitigate the consequences of an accident and is, therefore, reportable per 10 CFR 50.73(a)(2)(v)(D).

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Edwin I. Hatch Nuclear Plant Unit 2	☒ 050 ☐ 052	00366	2025	002	00

NARRATIVE

CORRECTIVE ACTIONS

The EOS was replaced and the 2A EDG successfully completed its operability surveillance on 2/15/2025. An extent-of-condition was conducted for the remaining EDGs. Two additional retaining nuts were found loose and subsequently corrected. Although two retaining nuts were found loose, the EOS's were installed in a position that would not result in the inoperablity of the EDGs. Additionally, procedures will be revised to capture installation guidance for the EOS.

PREVIOUS SIMILAR EVENTS

None.